

Question 33 (a)

Criteria	Marks
• Provides correct solution	2
• Finds the anti-derivative of $\frac{Ax}{x^2+4}$, or equivalent merit	1

Sample answer:

$$\int_0^6 \frac{Ax}{x^2+4} dx = 1$$

$$\frac{A}{2} \left[\ln(x^2+4) \right]_0^6 = 1$$

$$\frac{A}{2} (\ln 40 - \ln 4) = 1$$

$$\frac{A}{2} \left(\ln \frac{40}{4} \right) = 1$$

$$\frac{A}{2} \ln 10 = 1$$

$$A = \frac{2}{\ln 10}$$

Question 33 (b)

Criteria	Marks
• Provides correct solution	2
• Finds the correct derivative of $f(x)$, or equivalent merit	1

Sample answer:

$$f'(x) = \frac{A(x^2+4) - Ax(2x)}{(x^2+4)^2}$$

$$= \frac{Ax^2 + 4A - 2Ax^2}{(x^2+4)^2}$$

$$= \frac{4A - Ax^2}{(x^2+4)^2} = 0$$

$$\therefore 4A - Ax^2 = 0$$

When $Ax^2 = 4A$

$$x^2 = 4$$

$$x = 2, \quad x > 0$$

$\therefore$ The mode of X is 2.

Question 33 (c)

Criteria	Marks
• Provides correct solution	2
• Provides correct definite integral, or equivalent merit	1

Sample answer:

$$\begin{aligned}P(X < 2) &= \int_0^2 \frac{Ax}{x^2 + 4} dx \\&= \frac{A}{2} [\ln(x^2 + 4)]_0^2 \\&= \frac{A}{2} (\ln 8 - \ln 4) \\&= \frac{A}{2} \ln 2 \\&= \frac{2}{\ln 10} \times \frac{1}{2} \times \ln 2 \\&= \frac{\ln 2}{\ln 10} \\&= \log_{10} 2\end{aligned}$$

Question 33 (d)

Criteria	Marks
• Provides correct solution	2
• Attempts to find the conditional probability, or equivalent merit	1

Sample answer:

$$P(\text{IQ} > 130)$$

$$= P\left(\frac{\text{IQ} - 100}{15} > \frac{130 - 100}{15}\right)$$

$$= P(Z > 2)$$

$$= 2.5\%$$

$$P\left(\text{IQ} > 130 \mid \begin{array}{l} \text{Time to} \\ \text{complete} \end{array} < 2\right)$$

$$= \frac{P\left(\text{IQ} > 130 \cap \begin{array}{l} \text{Time to} \\ \text{complete} \end{array} < 2\right)}{P\left(\begin{array}{l} \text{Time to} \\ \text{complete} \end{array} < 2\right)}$$

$$= \frac{2.5\% \times 80\%}{\log_{10} 2}$$

$$= 0.06643\dots$$

$$= 0.066 \text{ (three decimal places)}$$